BAHIR DAR UNIVERSITY

BAHIR DAR INSTITUTE OF TECHNOLOGY

FACULTY OF COMPUTING

INDIVIDUAL PROJECT WORK

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DEP INFORMATION SYSTEMS

Course: Operating Systems

Project: Project Task 1

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Virtualization Tool Used: Oracle VM VirtualBox

1.Introduction

1.1 Background

An operating system (OS) is the most essential system software that acts as an interface between computer hardware and the user. It manages hardware resources such as CPU, memory, storage devices, and input/output peripherals while providing services for application programs. Without an operating system, a computer system cannot function effectively.

With the rapid growth of computing technology, modern systems increasingly depend on virtualization. Virtualization allows multiple operating systems to run simultaneously on a single physical machine by abstracting hardware resources. This has become a fundamental concept in data centers, cloud computing, server management, and software testing environments.

Arch Linux is a lightweight, flexible, and independently developed GNU/Linux distribution that follows a rolling-release model. Unlike other Linux distributions that provide graphical installers and preconfigured systems, Arch Linux focuses on simplicity, transparency, and user control. Because of this, it is highly valued in server environments and by system administrators who require deep understanding and customization of their systems.

1.2 Motivation

The motivation behind this project is to gain hands-on, real-world experience in installing and configuring an operating system inside a virtual environment. Virtual machines allow experimentation without the risk of damaging physical hardware, which is especially important for learning advanced operating system concepts.

Choosing Arch Linux Server provides an opportunity to:

Understand Linux internals in depth.Learn manual installation and configuration

Gain confidence in filesystem management and boot processes

Strengthen technical and problem-solving skills

This project bridges the gap between theoretical knowledge learned in class and practical system administration skills required in real computing environments.

2. Objectives

The main objectives of this project are:

To demonstrate understanding of operating system installation using virtualization tools

To install and configure Arch Linux Server in a virtual machine

To analyze hardware and software requirements for OS deployment

To gain practical experience in disk partitioning and filesystem selection

To identify problems faced during installation and apply appropriate solutions

To explain virtualization concepts (what, why, and how) clearly

To show confidence and expertise in operating system concepts

3. Requirements

3.1 Hardware Requirements (Host Machine)

To successfully install and run Arch Linux Server in a virtual environment, the host system must meet the following minimum hardware requirements:

Processor: Intel or AMD processor with virtualization support (VT-x or AMD-V enabled in BIOS)

RAM: Minimum 4 GB (8 GB recommended for smooth operation)

Storage: At least 20 GB free disk space

Network: Stable internet connection for package downloads

These requirements ensure that the virtual machine can operate efficiently without affecting the host system's performance.

3.2 Software Requirements

The following software components are required:

Host Operating System: Windows or Linux

Virtualization Software: Oracle VM VirtualBox

Guest OS: Arch Linux ISO (latest stable version)

VirtualBox was selected because it is free, open-source, widely supported, and suitable for academic learning environments.

4. Virtualization Environment

4.1 What is Virtualization?

Virtualization is a technology that enables multiple operating systems to run on a single physical machine by creating virtual versions of hardware resources. Each virtual machine behaves like a real computer, complete with its own CPU, memory, storage, and network interface.

4.2 Why Virtualization is Needed?

Virtualization is essential in modern operating systems because it provides:

Efficient utilization of hardware resources

Reduced hardware and maintenance costs

Safe environments for testing and learning

Server consolidation and scalability

Easy backup, recovery, and migration

4.3 How Virtualization Works?

Virtualization works through a hypervisor, which is software that manages and allocates physical resources to virtual machines. Oracle VM VirtualBox is a Type-2 hypervisor, meaning it runs on top of a host operating system.

The hypervisor:

Allocates CPU time to each VM

Assigns memory dynamically

Manages virtual disks and network interfaces

5. Virtual Machine Configuration

Type: Linux

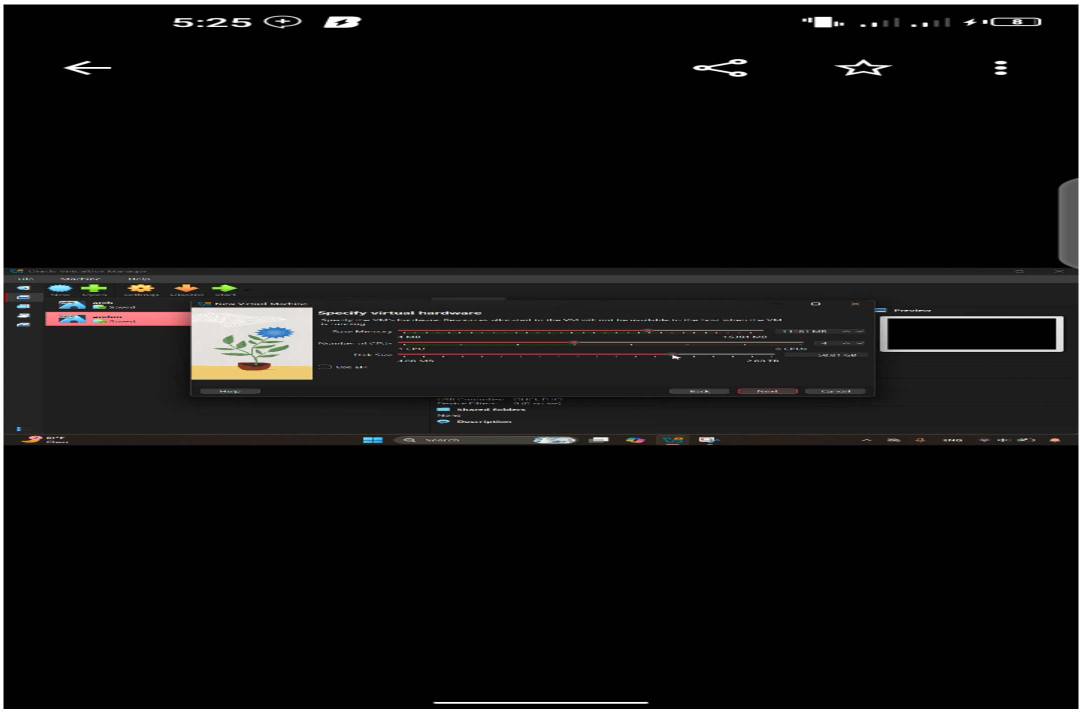
Version: Arch Linux (64-bit)

RAM: 2–4 GB

CPU: 1–2 cores

Disk Type: VDI (Dynamically Allocated)

Disk Size: 20 GB



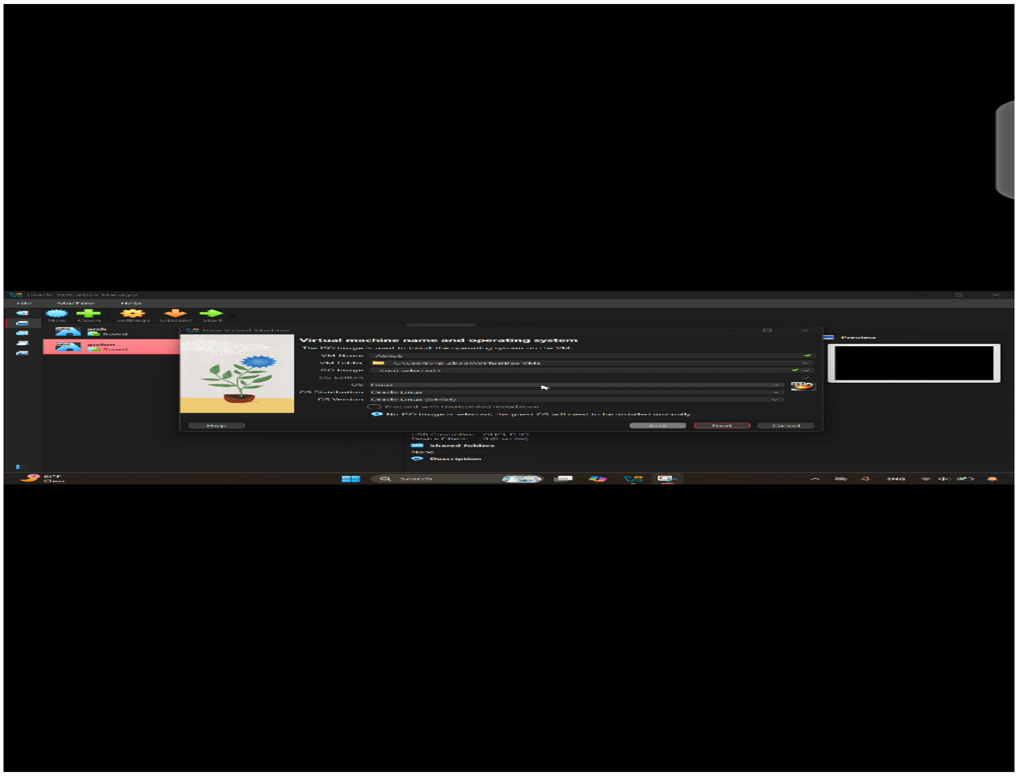
This configuration balances performance and efficient resource usage on the host machine.

6. Installation Steps of Arch Linux Server

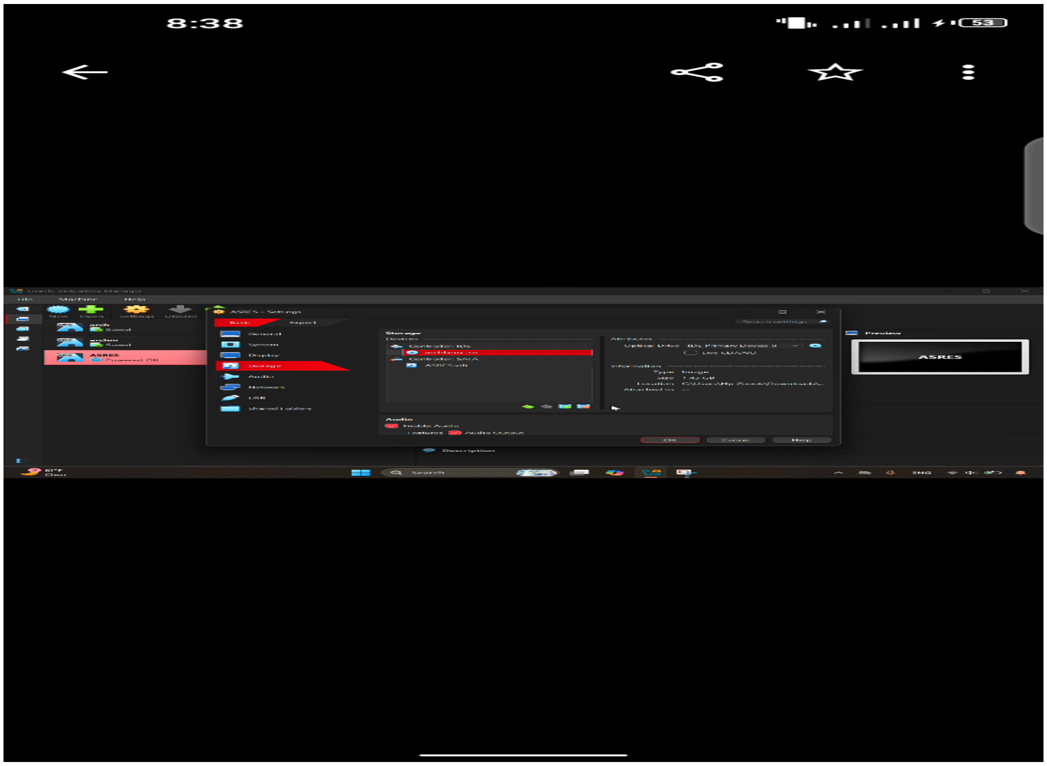
Important: Screenshots must be taken and inserted exactly at the locations marked below.

Step 1: Attach Arch Linux ISO

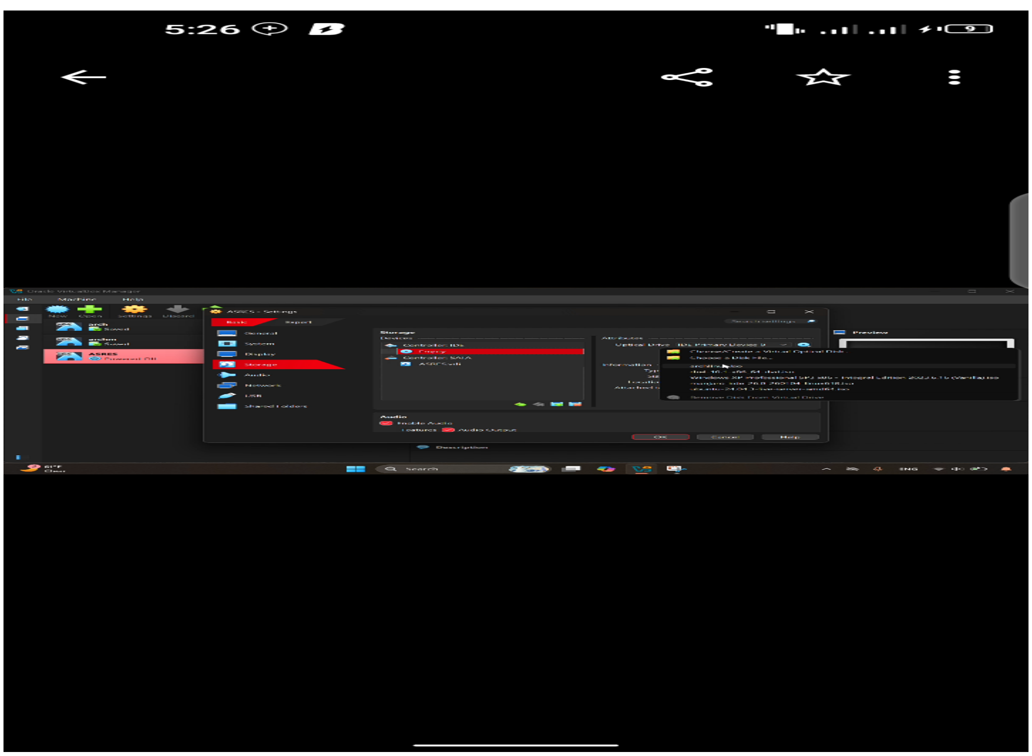
Open VirtualBox



Go to VM Settings → Storage



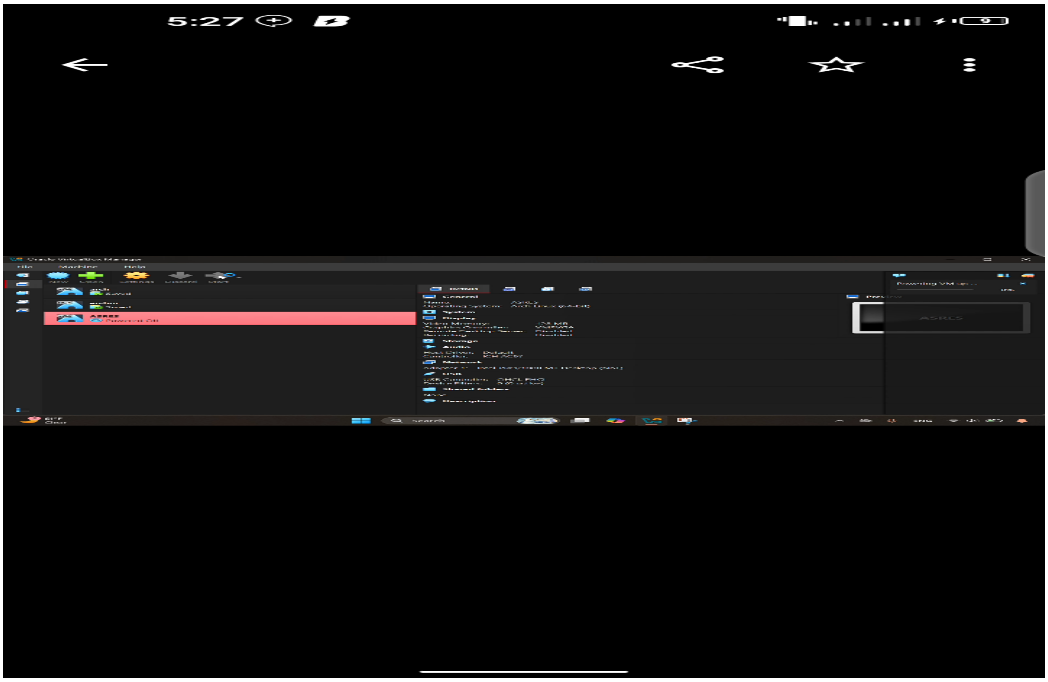
Attach Arch Linux ISO file



Step 2: Boot from Installation Media

Start the virtual machine

Select Arch Linux install medium



3: Internet Connection Check

Verify connectivity using:

ping archlinux.org

Step 4: Disk Partitioning

List disks using lsblk

Partition disk using cfdisk or fdisk

Create root (/) and optional swap partitions

Step 5: Filesystem Creation

Format root partition as ext4

mkfs.ext4 /dev/sda1

Step 6: Mount Partitions

Mount root partition:

mount /dev/sda1 /mnt

Step 7: Base System Installation

Install essential packages:

pacstrap /mnt base linux linux-firmware

Step 8: Generate fstab

Generate filesystem table:

genfstab -U /mnt >> /mnt/etc/fstab

Step 9: Chroot Environment

Enter system environment:

arch-chroot /mnt

Step 10: System Configuration

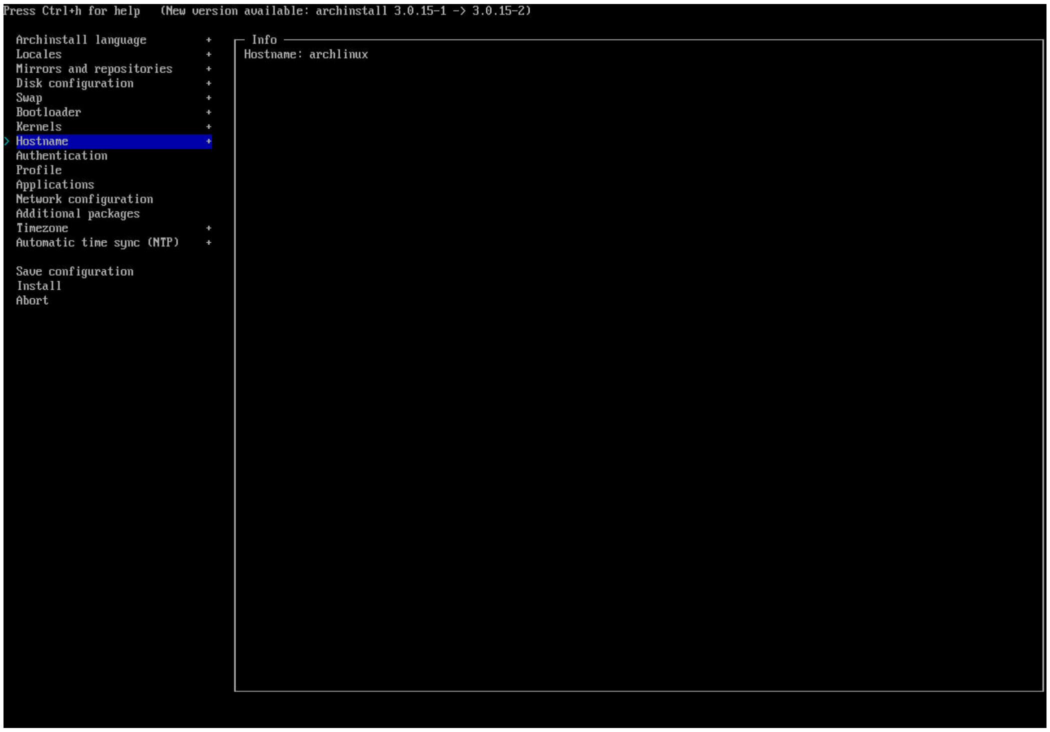
Configure timezone and locale

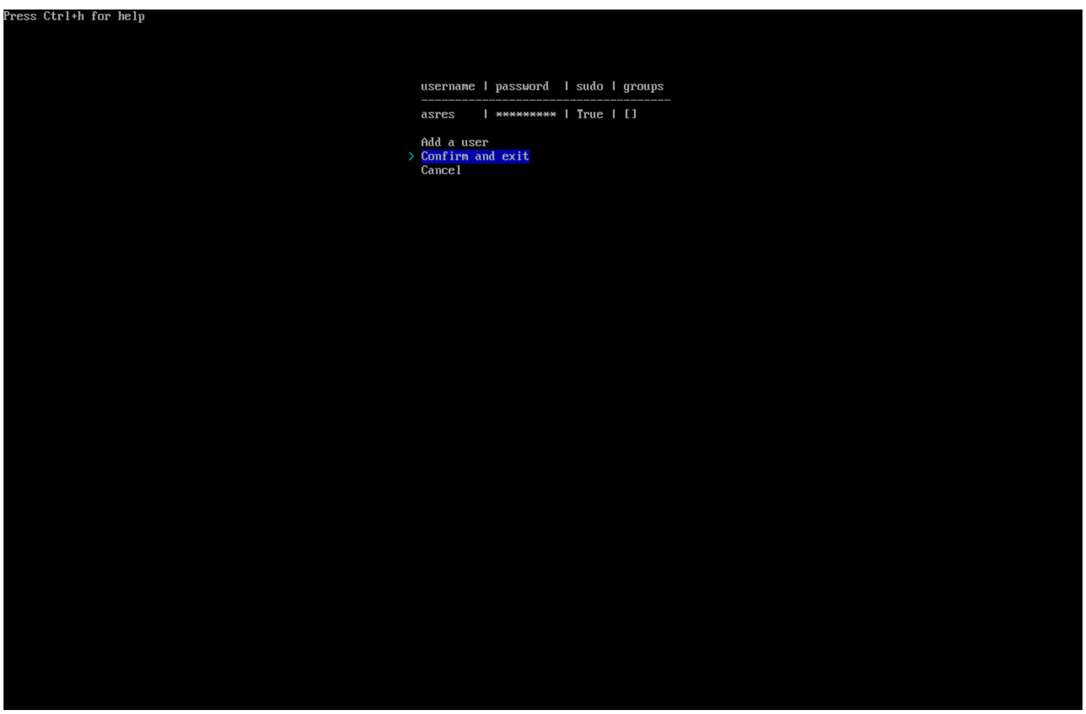
.Set hostname

Step 11: Bootloader Installation

Install and configure GRUD

Step 12: User Account Creation

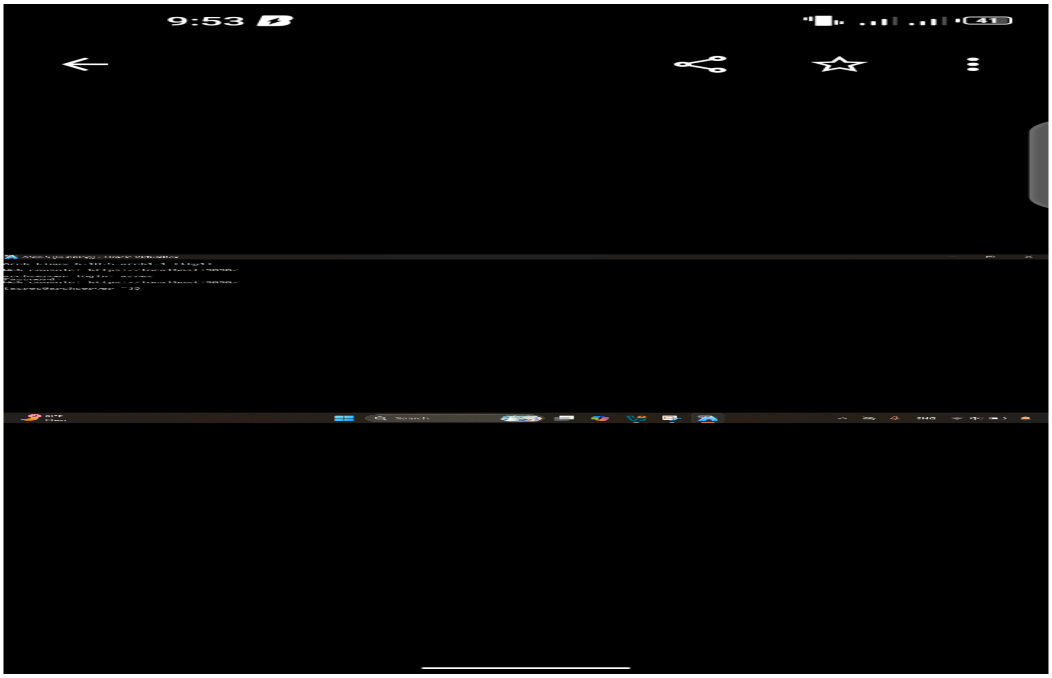




Assign password and sudo privileges

Step 13: Reboot System

Exit chroot



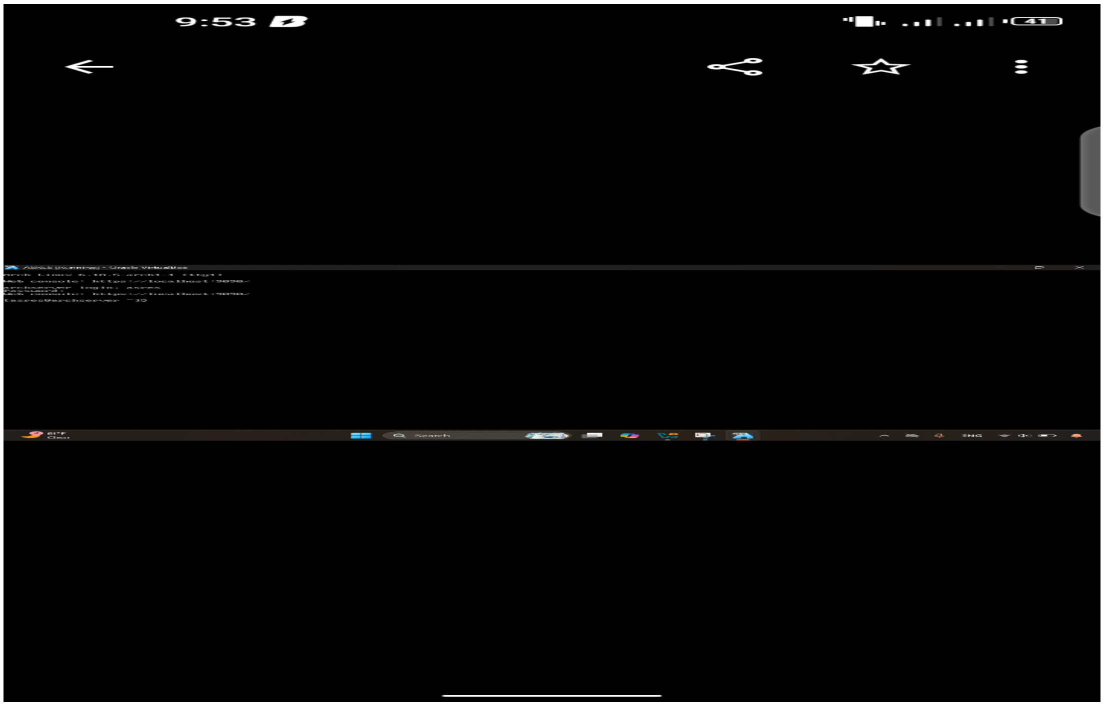
Unmount partitions

Reboot system

7. Issues (Problems Faced)

Issue 1: No Internet Connection

Issue 2: Bootloader Not Loading



8. Solutions

Enabled network adapter and DHCP

Reinstalled GRUB correctly

Verified disk names using lsblk

9. Filesystem Support

Supported Filesystems

NTFS – Windows

FAT32 – Compatibility

exFAT – Flash storage

ext4 – Linux servers

Btrfs – Snapshots

ZFS – Enterprise servers

HFS+ / APFS – macOS

Selected Filesystem: ext4

Reason: ext4 is stable, journaled, efficient, and best suited for Linux server environments.

10. Advantages and Disadvantages

Advantages

Lightweight and minimal

Rolling release

High customization

Disadvantages

Steep learning curve

Manual installation

11. Technical Evaluation (Practical Knowledge)

Windows → NTFS

Linux Server → ext4 / XFS

macOS → APFS

This demonstrates strong technical confidence and correct filesystem selection skills.

12. Conclusion

This project successfully demonstrated the installation and configuration of Arch Linux Server in a virtual environment. It strengthened understanding of operating systems, virtualization, and filesystem management.

13. Future Outlook / Recommendations

Explore advanced filesystems (Btrfs, ZFS)

Deploy server services (SSH, Web, Database)

Learn Docker and containerization

Practice automation and scripting